
8(a) (i) loss in E_K = gain in electric E_p M1

$$7.7 \times 10^{-13} = \frac{Qq}{4\pi\epsilon_0 r}$$
$$= \frac{(79 \times 1.60 \times 10^{-19})(2 \times 1.60 \times 10^{-19})}{4\pi(8.85 \times 10^{-12})r}$$

M1

$$r = 4.72 \times 10^{-14} \text{ m}$$

Since r is the distance of closest approach, the radius of gold must be less than this.

8(a) (ii) 1. most of the atom is empty space B1
or

the size of the nucleus is very small compared to the size of the atom

2. the nucleus is positively charged B1
the mass is concentrated in the nucleus B1

8(b)(i) ${}^0_{-1}\text{e}$ A1

